FUNDAMENTALS OF CYBER SECURITY
MODULE 5 — HANDWRITTEN QUESTIONS & ANSWERS

Exam-ready notes prepared from all 20 questions listed under Module 5 in the uploaded FCSDF Question Bank.

Q1. Explain the role of metadata analysis in digital forensics.

Answer:

Metadata is data about data. It provides information about a file, such as its name, size, type, creation time, modification time, author and sometimes location or device details.

Role of metadata analysis:

1. Establishes timelines: File creation and modification timestamps can help reconstruct events.
2. Identifies file ownership: Author and user details may help associate a file with an account.
3. Tracks file activity: Metadata may indicate when a file was created, opened, modified or accessed.
4. Supports authenticity checks: Metadata can help identify inconsistencies or signs of manipulation.
5. Links evidence: Similar timestamps, authors or device details can connect files and activities.
6. Helps identify applications: Software-specific metadata may indicate which application created a file.
7. Supports investigations: It can provide leads in cases involving fraud, data theft or unauthorized access.

Example:

In a data theft investigation, metadata shows that a confidential document was modified shortly before being copied to an external drive. Investigators can compare this information with USB logs and user activity.

Limitations:

Metadata can be changed, deleted or forged. Timestamps may also be affected by time zones and incorrect system clocks. Therefore, it should be corroborated with other evidence.

Conclusion: Metadata analysis helps establish context and timelines, but it should not be treated as conclusive proof on its own.

Q2. Describe how Windows Registry examination helps during a forensic investigation.

Answer:

The Windows Registry is a hierarchical database used by Windows to store configuration settings for the operating system, hardware, software and user accounts.

How Registry examination helps:

1. User activity: Can provide information about recently used applications and user settings.
2. USB history: Certain Registry keys can record details of previously connected USB devices.
3. Installed software: Helps identify applications installed on the computer.
4. System configuration: Provides details about operating system and hardware settings.
5. Login and user information: Some Registry artefacts can help identify user accounts and activity.
6. Network information: May contain details about previously used network configurations.
7. Persistence investigation: Certain locations may reveal programs configured to start automatically.

Important Registry locations:

- NTUSER.DAT: User-specific settings and activity artefacts.
- SYSTEM hive: System configuration and device-related information.
- SOFTWARE hive: Installed software and application configuration.
- SAM hive: Local account and security information, subject to access restrictions.

Example:

Investigators examine USB-related Registry artefacts to identify a removable storage device that was previously connected to a suspect computer. They correlate device identifiers and timestamps with other logs.

Conclusion: Registry examination helps reconstruct system and user activity, but findings should be verified using other forensic artefacts.

Q3. Differentiate between disk imaging and data recovery with suitable examples.

Answer:

Disk imaging and data recovery are two different processes used in digital investigations.

Disk Imaging:

- Creates a copy of a storage device, often a bit-by-bit forensic image.
- Main purpose is to preserve the original evidence for examination.
- Captures allocated and, where supported, unallocated areas.
- Commonly uses forensic imaging tools and write blockers.
- Example: Creating an image of a seized hard disk before examining it.

Data Recovery:

- Attempts to retrieve deleted, lost, corrupted or inaccessible data.
- Main purpose is to restore or extract available information.
- May use file-system analysis, carving and recovery utilities.
- Success depends on overwriting, encryption and damage.
- Example: Recovering a deleted document from a USB drive.

Key differences:

1. Purpose: Imaging preserves a copy; recovery retrieves lost or inaccessible data.
2. Output: Imaging produces an image file or clone; recovery produces recovered files or data.
3. Order: Investigators generally image first and perform recovery on a working copy.
4. Risk: Recovery attempts on original media may alter data, so they should be performed on a forensic copy whenever possible.

Conclusion: Disk imaging protects evidence, while data recovery attempts to retrieve information. Both are important in digital forensics.

Q4. Explain the purpose of Restore Points and Shadow Copies in retrieving system information.

Answer:

Restore Points and Shadow Copies are Windows features that can preserve earlier versions of system files or other data and may help investigators understand past system states.

Restore Points:

- Created by System Protection at selected times or before certain system changes.
- Store snapshots of important system files, Registry settings and configuration.
- Mainly help restore Windows system settings to an earlier state.
- They are not a complete backup of all personal files.

Shadow Copies:

- Created using the Volume Shadow Copy Service (VSS).
- May preserve previous versions of files and folders on a volume.
- Can help retrieve earlier versions of files if copies still exist.
- May provide useful historical information during forensic analysis.

Forensic uses:

1. Recover earlier versions of certain files.
2. Compare changes made to system configuration.
3. Identify files that existed before deletion or modification.
4. Support timeline reconstruction.
5. Examine earlier system states.

Example:

A user deletes a report. If a suitable Shadow Copy still exists, an investigator may examine an earlier version of the report from that snapshot.

Limitations:

Restore Points or Shadow Copies may be disabled, deleted, overwritten or unavailable. Their existence and timestamps must be verified.

Conclusion: These Windows features may provide historical system and file information, but they are not guaranteed backups or guaranteed recovery sources.

Q5. What is the significance of analyzing the Recycle Bin during digital forensic analysis?

Answer:

The Windows Recycle Bin is a temporary storage location where deleted files may be kept before permanent removal. Its examination can provide useful evidence about deleted user data.

Significance:

1. File recovery: Deleted files may be restored if they remain in the Recycle Bin.
2. Original file path: Metadata may preserve the original location of a deleted file.
3. Deletion time: Information may help establish when a file was deleted.
4. User activity: Deleted items can indicate actions performed by a user.
5. Evidence discovery: Relevant documents, images or other files may be found.
6. Timeline reconstruction: Deletion details can be correlated with other system artefacts.

Windows Recycle Bin artefacts:

- $I files: Store metadata such as original file path, deletion time and original file size.
- $R files: Contain the deleted file's actual data while it remains in the Recycle Bin.

Example:

During an insider data theft investigation, an examiner finds a deleted spreadsheet in the Recycle Bin. The $I metadata identifies its original path and deletion time, while the matching $R file contains the spreadsheet.

Limitations:

If the Recycle Bin has been emptied, the files may no longer be available there. Other recovery methods may be required.

Conclusion: Recycle Bin analysis can reveal deleted files and related metadata, helping investigators understand user activity.

Q6. Illustrate the function of EnCase and FTK tools in forensic investigations.

Answer:

EnCase and FTK (Forensic Toolkit) are digital forensic tools used to acquire, examine, search and analyse digital evidence.

1. EnCase:

- Creates forensic images of storage devices.
- Supports evidence preservation and integrity verification.
- Examines files, folders, partitions and file-system artefacts.

- Supports recovery and analysis of deleted files.
- Provides search, filtering and reporting features.
- Helps maintain case and evidence documentation.

2. FTK (Forensic Toolkit):

- Acquires and processes digital evidence.
- Indexes files to support fast keyword searching.
- Examines emails, documents, images and other file types.
- Supports recovery and analysis of deleted data.
- Provides filtering, bookmarking and reporting features.
- Helps organize evidence for investigations.

Example:

In a corporate data theft case, an investigator creates a forensic image of a suspect's hard disk. The examiner uses EnCase or FTK to search for confidential documents, inspect metadata and identify relevant user activity.

Conclusion: EnCase and FTK help investigators handle and analyse large amounts of digital evidence in a structured manner. Correct procedures and verification are still essential.

Q7. Explain anti-forensics techniques and their impact on digital investigations.

Answer:

Anti-forensics refers to techniques used to hide, destroy, alter or make digital evidence difficult to find or analyse.

Common techniques:

1. Secure deletion: Overwriting files to make recovery difficult.
2. Data wiping: Erasing storage areas or entire devices.
3. Encryption: Making data unreadable without the correct key.
4. Steganography: Hiding information inside images, audio or other files.
5. Log tampering: Deleting or modifying system and application logs.
6. Timestamp manipulation: Changing file timestamps to mislead investigators.
7. File hiding: Changing attributes, extensions or locations to conceal files.
8. Evidence obfuscation: Using misleading names, formats or fragmented data.

Impact on investigations:

- Evidence may become unavailable or difficult to recover.
- Timelines may become unreliable.
- Investigations may take longer and cost more.
- Important logs and user activity may be missing.
- Investigators may face difficulty proving authenticity and attribution.

Countermeasures:

- Create forensic images as early as possible.
- Preserve volatile data when necessary.
- Compare multiple evidence sources.
- Examine file-system artefacts and backups.
- Use forensic tools to detect inconsistencies.
- Verify evidence with cryptographic hashes.
- Document limitations and uncertainty.

Conclusion: Anti-forensics attempts to obstruct investigations. Multiple evidence sources and careful forensic procedures can help identify or reduce its impact.

Q8. Describe vulnerability assessment tools and how they support digital forensics.

Answer:

Vulnerability assessment tools scan systems, networks and applications to identify known security weaknesses, misconfigurations and outdated software.

Examples of tools:

1. Nessus: Scans systems for known vulnerabilities and configuration issues.
2. OpenVAS / Greenbone: Performs vulnerability scanning and generates reports.
3. Nmap: Discovers hosts, open ports and running services; it can support security assessment.
4. Qualys: Provides vulnerability and configuration assessment services.
5. Nikto: Checks web servers for known issues and potentially unsafe configurations.

How they support digital forensics:

- Identify weaknesses that may have been exploited during an attack.

- Help understand possible attack paths.
- Provide information about exposed services and affected systems.
- Support incident investigation and remediation.
- Help correlate vulnerabilities with logs and observed attack activity.

Example:

After a server compromise, investigators find an outdated web service. A vulnerability assessment identifies a known weakness in that service. Investigators compare this finding with server logs and other evidence to assess whether it may have been exploited.

Note: Vulnerability scanners identify potential weaknesses; they do not by themselves prove that an attacker exploited them.

Conclusion: These tools provide technical context for investigations and help organizations identify and fix weaknesses.

Q9. Explain the steps involved in examining tampered/deleted data.

Answer:

Examining tampered or deleted data means identifying possible modifications or deletion and attempting to recover or interpret available evidence.

Steps:

1. Secure the device: Obtain authorization and prevent further changes.
2. Document evidence: Record device details, condition and chain of custody.
3. Create forensic image: Use a write blocker where applicable and preserve the original.
4. Verify integrity: Calculate and record cryptographic hashes.
5. Examine file systems: Review partitions, directories, allocation records and metadata.
6. Identify suspicious changes: Look for unusual timestamps, missing files and altered metadata.
7. Recover deleted data: Use file-system recovery and file carving on the forensic copy.
8. Examine logs and artefacts: Review system logs, application records and user activity.
9. Compare versions: Use backups, Shadow Copies or other sources where

available.

10. Document results: Record recovered data, methods, limitations and hash values.

Example:

A company suspects that an employee deleted confidential files. Investigators image the disk, examine file-system records and search for recoverable files. They compare timestamps with USB connection records and system logs.

Conclusion: A careful process can recover or identify some deleted or altered data while preserving the original evidence.

Q10. Discuss the role of data recovery tools in retrieving user information from storage devices.

Answer:

Data recovery tools are software applications used to retrieve deleted, lost, corrupted or inaccessible information from storage devices.

Role of data recovery tools:

1. Recover deleted files: Retrieve files that have not been overwritten.
2. Restore lost partitions: Help locate or rebuild partition information in suitable cases.
3. Recover damaged file systems: Extract data from certain corrupted file systems.
4. File carving: Identify files using known file signatures when directory records are missing.
5. Retrieve user information: Recover documents, photos, videos and other user files.
6. Support investigations: Help locate evidence relevant to fraud, data theft or other incidents.

Examples:

- PhotoRec: File-carving tool that can recover files based on signatures.
- TestDisk: Helps analyse and recover lost partitions and repair certain disk structures.
- Recuva: General-purpose file recovery utility.
- Autopsy / The Sleuth Kit: Forensic platform and tools for examining disk images and file systems.

Important precautions:

- Avoid writing new data to the source device.
- Create a forensic image before recovery where possible.
- Work on a copy, not the original evidence.
- Record tools, settings and recovery results.
- Verify and document recovered evidence.

Conclusion: Data recovery tools help retrieve information, but recovery is not guaranteed, especially when data has been overwritten or securely erased.

Q11. Analyze how metadata can be used to trace file activity in a cybercrime case.

Answer:

Metadata contains details about a file, including its name, size, type, timestamps, author and application information. It can help investigators trace file activity.

How metadata helps:

1. Creation time: Indicates when a file may have been created.
2. Modification time: Helps identify when content was last changed.
3. Access time: May indicate when a file was accessed, depending on system settings.
4. Author details: May identify an account or application associated with a document.
5. File path: Helps identify where the file was stored.
6. Device or software details: May provide information about the source device or application.
7. Correlation: Metadata can be compared with logs, emails and USB records.

Case example:

A confidential file is found on a USB drive. Its metadata shows a modification time shortly before the USB device was connected. Investigators compare the file details with USB history, login records and access logs to build a possible timeline.

Limitations:

Timestamps can be changed, copied or affected by time-zone settings. Some systems do not update access times reliably. Metadata alone cannot prove who performed an action.

Conclusion: Metadata is useful for tracing file activity, but it must be correlated with independent evidence.

Q12. Demonstrate how a forensic investigator can retrieve deleted files using Restore Points and Shadow Copies.

Answer:

Windows Restore Points and Shadow Copies may preserve earlier system or file versions. An investigator can examine them to locate information that was deleted or modified.

Procedure:

1. Obtain proper authorization and secure the computer or forensic image.
2. Document the system condition and preserve evidence.
3. Create a forensic image before conducting recovery attempts.
4. Identify available restore points and Volume Shadow Copies.
5. Record snapshot dates, times and identifiers.
6. Mount or access the relevant snapshot using suitable forensic methods.
7. Browse the previous file and folder versions.
8. Identify the required deleted or modified file.
9. Export the file to a separate evidence location.
10. Calculate a SHA-256 hash and record the recovery details.
11. Compare the recovered file with other evidence and document findings.

Example:

A user deletes a project report on Monday. A Shadow Copy created on Sunday may contain an earlier version of the report. The examiner extracts it from the snapshot and records its hash.

Important points:

- Restore Points mainly protect system settings and are not a complete backup of personal files.
- Shadow Copies may not exist or may have been deleted.
- Recovery depends on snapshot availability and retention.
- All recovery should be performed on a forensic copy wherever possible.

Conclusion: Restore Points and Shadow Copies can provide earlier versions of data, but recovery depends on whether suitable snapshots still exist.

Q13. Apply the procedure of Windows Registry analysis to track USB usage history.

Answer:

Windows Registry analysis can help identify USB storage devices previously connected to a computer. It may reveal device identifiers and other connection-related information.

Procedure:

1. Obtain authorization and create a forensic image of the system.
2. Identify and preserve relevant Registry hive files, especially the SYSTEM hive.
3. Load the Registry hive using a suitable forensic Registry viewer.
4. Examine the USBSTOR Registry location:
SYSTEM\CurrentControlSet\Enum\USBSTOR
5. Record device vendor, product and instance identifiers where available.
6. Examine related device and mount information, including MountedDevices and relevant Enum entries.
7. Correlate the findings with setup logs, event logs and user activity artefacts.
8. Record timestamps only where the source artefact supports them.
9. Document the tools, Registry paths and findings.

Example:

Investigators find a USB device identifier in the USBSTOR Registry key. They compare the device details with a file-copy timeline and other logs to assess whether the device may have been used during a data theft incident.

Limitations:

Registry artefacts may be incomplete, overwritten or affected by Windows version differences. A Registry entry can show prior device presence but does not by itself prove which files were copied.

Conclusion: Registry analysis helps identify USB device history and should be combined with other evidence for reliable conclusions.

Q14. Compare the features and capabilities of EnCase and FTK tools in forensic investigations.

Answer:

EnCase and FTK are digital forensic tools used to acquire, process,

examine and report digital evidence.

EnCase:

- Supports forensic acquisition and imaging.
- Provides file-system and disk analysis.
- Supports recovery and examination of deleted files.
- Offers searching, filtering and evidence reporting.
- Provides case management and evidence documentation.

FTK:

- Supports forensic acquisition and evidence processing.
- Uses indexing to support fast keyword searches.
- Examines emails, documents and many file types.
- Supports deleted file analysis and recovery.
- Offers filtering, bookmarking and reporting features.

Comparison:

1. Main strength: EnCase is widely known for forensic acquisition and examination workflows; FTK is known for indexing and searching large evidence collections.
2. Evidence analysis: Both support file-system and artefact examination.
3. Search: FTK emphasizes indexed searching; EnCase also provides search and filtering capabilities.
4. Reporting: Both provide reporting and case documentation.
5. Usage: Selection depends on investigation needs, licensing, examiner experience and supported evidence formats.

Conclusion: Both tools support professional forensic investigations. Neither tool replaces correct evidence handling, examiner judgement or validation of findings.

Q15. Analyze a scenario where the Recycle Bin is emptied intentionally—identify possible forensic recovery methods.

Answer:

Scenario:

An employee intentionally deletes confidential company files and empties the Windows Recycle Bin to hide evidence.

Possible forensic methods:

1. Preserve the device: Stop unnecessary use to reduce the chance of

overwriting deleted data.

2. Create a forensic image: Acquire the storage device using appropriate forensic procedures.

3. Examine file-system records: Check for deleted directory entries and allocation information.

4. Perform file carving: Search unallocated space for known file signatures.

5. Examine Shadow Copies: Check whether previous versions of files are available.

6. Check backups: Review approved backup systems and archived copies.

7. Examine application artefacts: Search recent-file lists, temporary files and application caches.

8. Review logs: Correlate file access, user logins and USB device activity.

9. Examine cloud or network copies: Where authorized, review synchronized folders or server-side versions.

10. Document limitations: Note whether files were overwritten, encrypted or securely erased.

Tools:

Autopsy, The Sleuth Kit, EnCase, FTK and PhotoRec may support relevant examination or recovery tasks.

Conclusion:

Emptying the Recycle Bin does not always permanently remove all traces. Recovery may be possible if data or related artefacts remain, but success is not guaranteed.

Q16. Explain how disk imaging can support the chain of custody in digital investigations.

Answer:

Disk imaging creates a forensic copy of a storage device. It supports chain of custody by preserving the original evidence and allowing examination to be performed on a copy.

How it supports chain of custody:

1. Original preservation: The source device can be stored securely after acquisition.

2. Documentation: Imaging records can include device details, date, time, examiner and tool information.

3. Integrity verification: Hash values can help detect changes to the acquired image.
4. Repeatable examination: Multiple examiners can work from a verified copy.
5. Reduced handling: The original device does not need to be repeatedly accessed.
6. Evidence tracking: Image files can be labelled with case and evidence identifiers.
7. Transfer verification: Hashes can be recalculated after copying or transfer.

Procedure:
Identify device → Document details → Use write blocker where applicable → Create forensic image → Calculate hashes → Verify image → Seal and store original → Examine working copy.

Example:
A seized hard disk is imaged and its SHA-256 hash is recorded. The image is transferred to the forensic laboratory. The examiner recalculates the hash and confirms it matches the recorded value.

Conclusion: Disk imaging, combined with documentation, hashing and secure storage, helps preserve evidence integrity and supports a reliable chain of custody.

Q17. Evaluate different anti-forensic methods and propose counter-techniques to overcome them.

Answer:
Anti-forensic methods are techniques used to hide, destroy or manipulate digital evidence to obstruct an investigation.

Methods and counter-techniques:

1. Secure deletion / wiping:
Impact: Makes deleted data difficult or impossible to recover.
Counter-technique: Examine backups, snapshots, logs and other devices for remaining copies.

2. Encryption:
Impact: Prevents access to file contents without valid keys.
Counter-technique: Preserve key material lawfully, examine available metadata and seek authorized access.

3. Steganography:
Impact: Hides information inside image, audio or video files.
Counter-technique: Use steganalysis and compare files with known originals.

4. Log deletion:
Impact: Removes records of system and user activity.
Counter-technique: Examine centralized logs, backups, network records and logs on other systems.

5. Timestamp manipulation:
Impact: Misleads investigators about when activity occurred.
Counter-technique: Correlate timestamps from independent sources and consider time-zone differences.

6. File hiding:
Impact: Makes files difficult to locate.
Counter-technique: Examine hidden files, unusual extensions, unallocated space and file-system structures.

7. Evidence alteration:
Impact: Reduces confidence in the authenticity of files.
Counter-technique: Compare hashes, backups, metadata and independent copies.

Conclusion: No single counter-technique defeats every anti-forensic method. Investigators should use multiple sources, preserve evidence early and clearly document uncertainty.

Q18. Given a tampered hard disk, outline the process of recovering maximum possible data using forensic tools.

Answer:
A tampered hard disk may contain deleted, modified, hidden or damaged information. The goal is to preserve the disk and recover as much available evidence as possible.

Procedure:
1. Secure the disk: Label it, document its condition and maintain chain of custody.
2. Avoid normal booting: Do not start the operating system from the suspect disk.

3. Inspect physical condition: Check for visible damage and use a suitable forensic workstation.
4. Create a forensic image: Use a write blocker where applicable and a trusted imaging tool.
5. Record hashes: Calculate and record cryptographic hash values for verification.
6. Handle read errors: Use suitable acquisition settings and document unreadable sectors.
7. Examine partitions and file systems: Identify lost partitions and damaged structures.
8. Recover deleted files: Use file-system recovery and file carving on the forensic copy.
9. Examine metadata and logs: Search for traces of file changes, user activity and system events.
10. Check backups and snapshots: Examine Shadow Copies and other authorized backup sources.
11. Search for hidden data: Inspect unusual files, alternate data streams and unallocated space where relevant.
12. Validate results: Compare recovered information with independent evidence.
13. Report findings: Record tools, methods, hashes, recovered files and limitations.

Possible tools:

FTK, EnCase, Autopsy, The Sleuth Kit, PhotoRec and TestDisk.

Conclusion: Imaging first, working on verified copies and combining recovery methods provides the best chance of retrieving available evidence without damaging the original.

Q19. Examine the role of vulnerability assessment tools in identifying cyber-attack sources during forensic analysis.

Answer:

Vulnerability assessment tools identify weaknesses in systems, applications and networks. During forensic analysis, their results can help investigators understand how an attacker may have gained access.

Role:

1. Identify exposed services: Tools such as Nmap can show open ports

and running services.

2. Detect known weaknesses: Vulnerability scanners can identify outdated software and known vulnerabilities.

3. Understand attack paths: Findings may indicate possible routes an attacker could have used.

4. Prioritize affected systems: Helps identify systems that may need deeper investigation.

5. Correlate evidence: Scanner results can be compared with firewall, server and intrusion logs.

6. Support remediation: Helps recommend patches and configuration changes.

Example:

A web server is compromised. A vulnerability scan identifies an outdated service with a known weakness. Investigators compare this result with access logs, network records and malware artefacts to assess whether it may have been involved in the incident.

Important limitation:

A vulnerability assessment identifies potential weaknesses. It does not independently identify the attacker or prove that a particular vulnerability was exploited.

Conclusion: Vulnerability assessment tools provide technical context and help investigators examine possible attack paths when combined with forensic evidence.

Q20. Investigate a case where partial data wiping is performed and recommend the best data recovery tools for evidence retrieval.

Answer:

Scenario:

An attacker partially wipes a hard disk to destroy selected files while leaving other data intact. Investigators must preserve the disk and attempt to retrieve remaining evidence.

Investigation procedure:

1. Secure and document the disk.

2. Do not write new data to the source device.

3. Create a forensic image using a suitable acquisition tool.

4. Record and verify hash values.
5. Examine partitions, file systems and unallocated space.
6. Identify wiped areas and possible remaining file-system records.
7. Use file carving to search for recoverable file contents.
8. Examine backups, Shadow Copies and synchronized copies where authorized.
9. Correlate recovered files with metadata, logs and other devices.
10. Document recovered data, damaged areas and limitations.

Recommended tools:

- PhotoRec: Useful for file carving based on file signatures.
- TestDisk: Useful for examining and recovering lost partitions and certain file-system structures.
- Autopsy / The Sleuth Kit: Useful for forensic examination of disk images and file-system artefacts.
- EnCase: Supports forensic acquisition, examination and reporting.
- FTK: Supports processing, indexing, searching and examination of evidence.

Tool selection:

PhotoRec is useful when file-system records are missing but file contents remain. TestDisk may help when partition structures are damaged. Autopsy, EnCase or FTK can support broader forensic examination.

Limitations:

If sectors have been overwritten or securely erased, recovery may be impossible. SSD TRIM and device behaviour can also reduce recovery chances.

Conclusion: The best approach is to image the disk first, then combine file-system analysis, file carving and backup examination. No tool can guarantee recovery of overwritten data.